

Analysis of gender differences in the proportion of sad or hopeless adolescents.

Group & Response Definitions

Group Definition (Independent Variable):

Female =1 Male = 0

Response Variable (Dependent Variable):

Yes (=1): Felt sad or hopeless

No (=0): Did not feel sad or hopeless

Method

Statistical Method: Two-sample z-test for proportions (Two-tailed test).

Hypotheses:

Null Hypothesis (H0): $P_{\text{female}} = P_{\text{male}}$

Alternative Hypothesis (H1): $P_{\text{female}} \neq P_{\text{male}}$

Conclusion

Main Finding:

1. Gender Difference: Rates of feeling sad or hopeless differ significantly by gender.
2. Higher in Females: The proportion is significantly higher among female students.

Key Results

Sample Proportions:

Female: 37.18% Male: 22.73%
reported feeling sad or hopeless.

Estimated Difference:

The proportion of females is 14.44 percentage points higher than males

Significance Test:

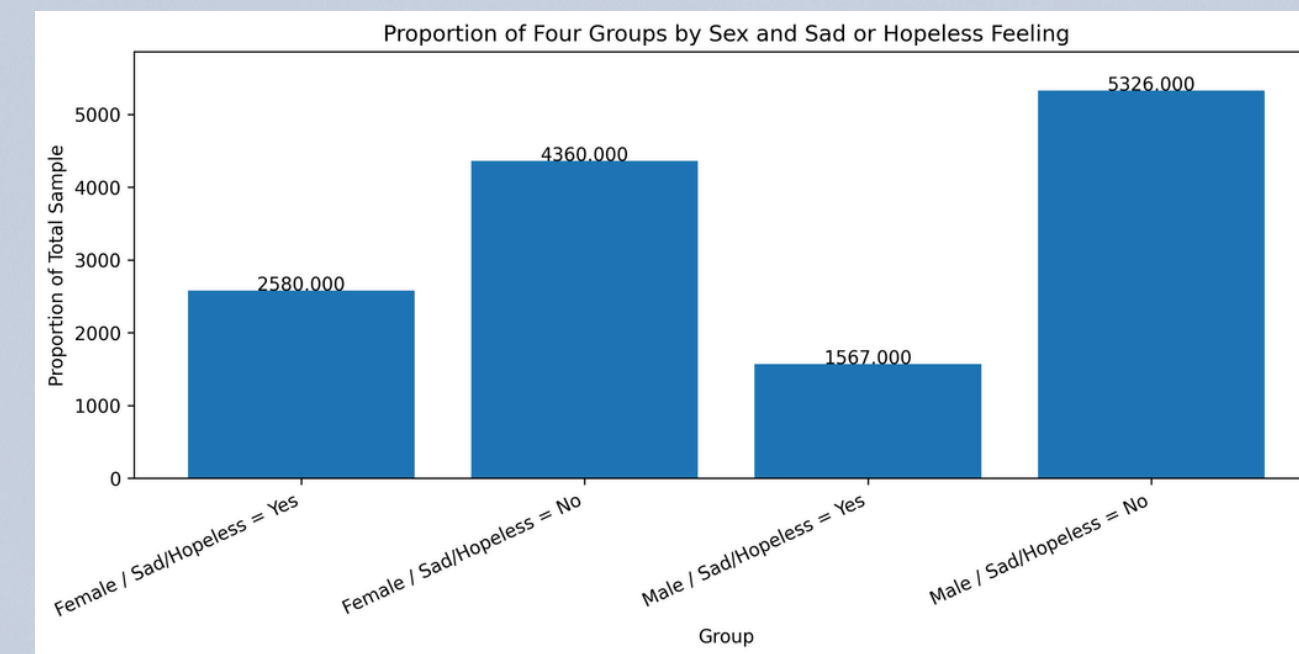
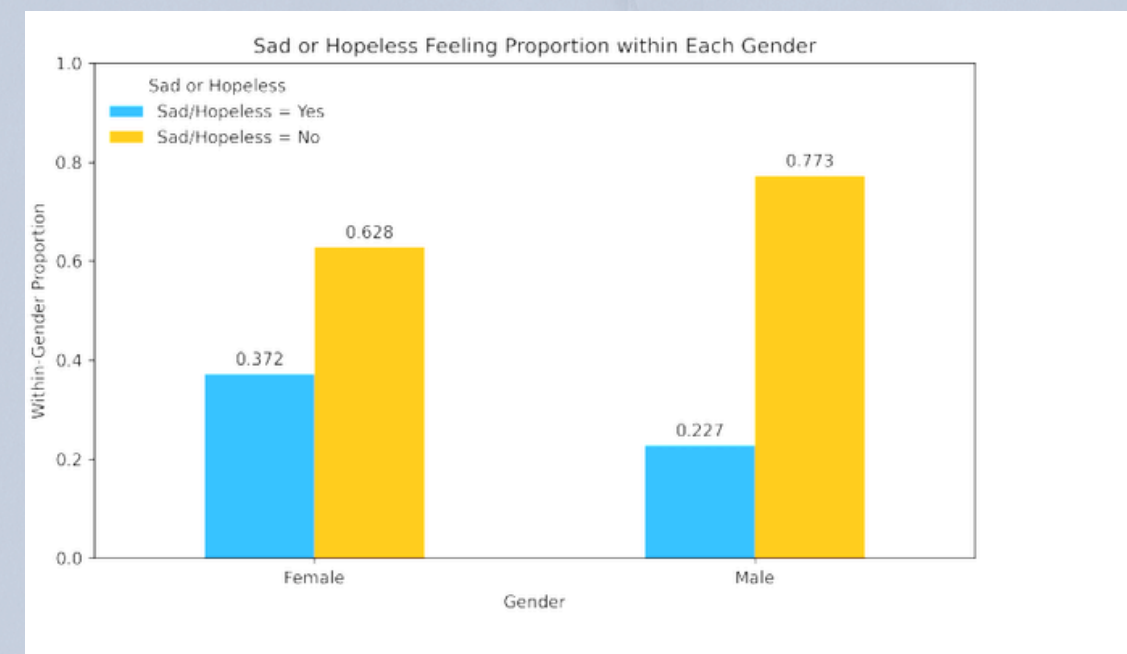
At a significance level of $\alpha = 0.05$, the p-value = 0.0000 ($p < 0.05$).

Decision: Reject the Null Hypothesis (H0).

95% Confidence Interval:

The 95% confidence interval for the difference is [0.1294, 0.1595] (or 12.94% to 15.95%).

Bar Chart



Females report a significantly higher rate of feeling sad or hopeless (37.2%) compared to males (22.7%).